Day 3 — Types of Generative AI Models

1️⃣ What are Large Language Models?

Large Language Models (LLMs) are advanced **deep learning models** trained on massive amounts of text data to understand, generate, and manipulate human language.

They can perform many natural language tasks such as:

- Text generation
- Question answering
- Translation
- Summarization
- Chatbots
- Code generation

LLMs learn **patterns, grammar, semantics, and reasoning** from huge datasets like:

- books
- websites
- research papers
- code repositories
- articles

The word **"Large"** means:

- billions or trillions of parameters
- trained on extremely large datasets
- require huge compute resources.

Examples include models from organizations like **OpenAI**, **Google DeepMind**, and **Anthropic**.

2️⃣ Core Idea Behind LLMs

The main idea of LLMs is **next-word prediction**.

The model learns to predict the **next token in a sequence**.

Example:

Input:

AI will change the

Prediction:

world

During training the model repeatedly learns patterns like this from billions of sentences.

Over time it learns:

- grammar
- context
- relationships between words
- reasoning patterns.

3️⃣ Architecture Used in LLMs

Modern LLMs are built using the **Transformer architecture**, introduced in the famous paper **"Attention Is All You Need" (2017)**.

Key components:

1. Tokenization
2. Embedding layer
3. Positional encoding
4. Transformer layers
5. Output prediction layer

Simple LLM Pipeline Diagram

User Input

↓

Tokenization

↓

Embedding Layer

↓

Transformer Layers (Self Attention + Feed Forward)

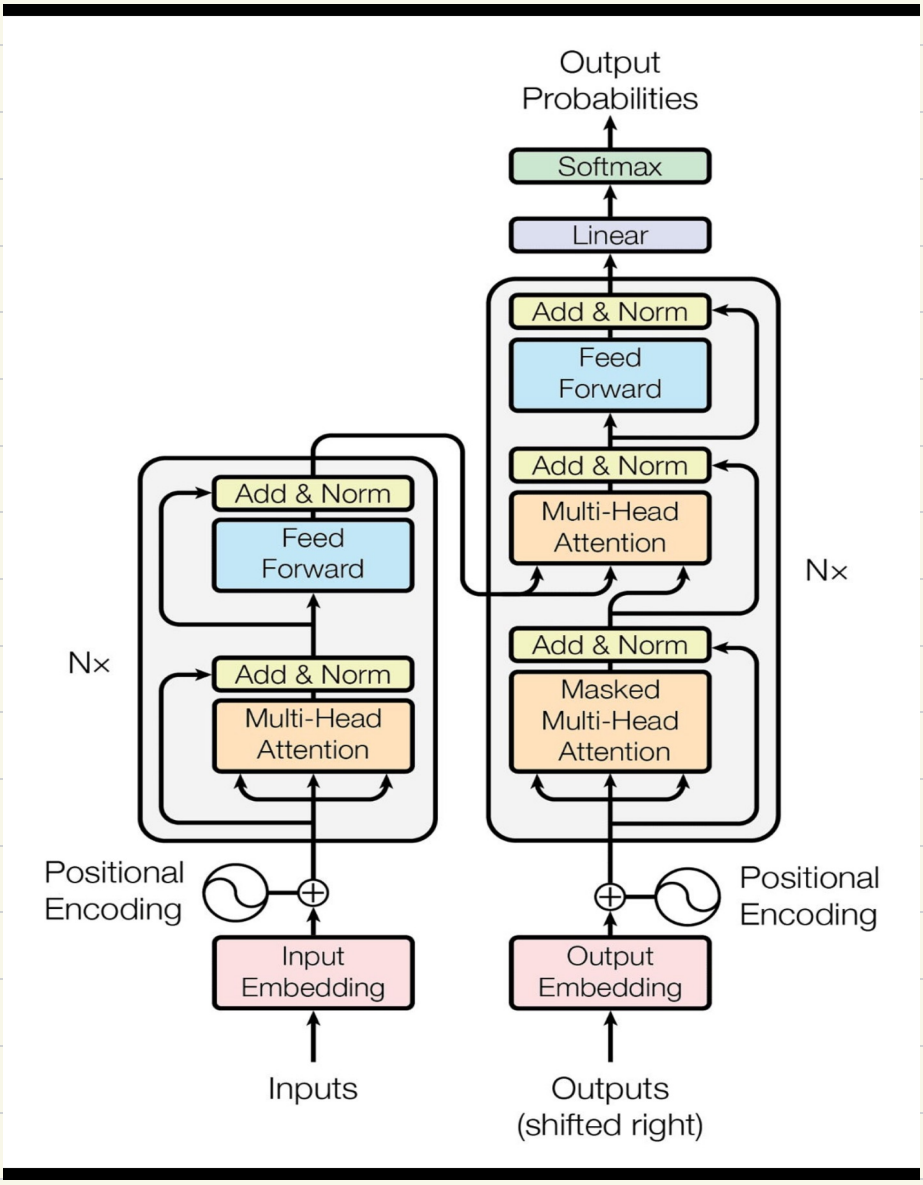
↓

Output Probability

↓

Generated Text

Transformer



2️⃣ Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are a **type of generative deep learning model** where **two neural networks are trained together in a competitive process** to generate new data that looks similar to real data.

GANs were introduced in **2014** by **Ian Goodfellow**.

The two networks are:

1️⃣ **Generator (G)** – generates fake data
2️⃣ **Discriminator (D)** – identifies whether the data is real or fake

The goal of GAN training is:

- The **Generator tries to fool the Discriminator**
- The **Discriminator tries to detect fake data**

Through this competition, the generator eventually learns to produce **very realistic synthetic data**.

2. Core Idea of GAN (Very Simple Explanation)

Think of GAN like a **game between two players**.

Player	Role
Generator	Creates fake data
Discriminator	Detects fake data

Example:

Generator creates **fake human faces**.
Discriminator checks if the face is **real or generated**.

Over time:

- Generator becomes **better at generating**
- Discriminator becomes **better at detecting**

Finally the generated data becomes **almost indistinguishable from real data**.

GAN Training Process (Step-by-Step)

How GAN Works (Training Process)

Step 1 — Generate Noise

- Generator takes **random noise vector (z)** as input.
- Noise usually comes from **Gaussian or Uniform distribution**.

```code
Noise (z) → Generator
```

Step 2 — Generate Fake Data

- The **Generator (G)** converts noise into **fake data**.

Examples:

- fake images
- fake audio
- fake text

```code
Noise → Generator → Fake Data
```

Step 3 — Discriminator Receives Data

The **Discriminator (D)** receives two inputs:

1. **Real data** (from dataset)
2. **Fake data** (from generator)

```code
Real Data → Discriminator
Fake Data → Discriminator
```

Step 4 — Discriminator Classification

The discriminator predicts whether the data is **real or fake**.

Output:

```code
Probability (Real vs Fake)
```

Example:

```code
0.95 → Real
0.05 → Fake
```

Step 5 — Update Discriminator

The discriminator learns to **better distinguish real and fake data**.

Goal:

```code
Correctly classify real and fake samples
```

Step 6 — Update Generator

The generator updates its parameters to **fool the discriminator**.

Goal:

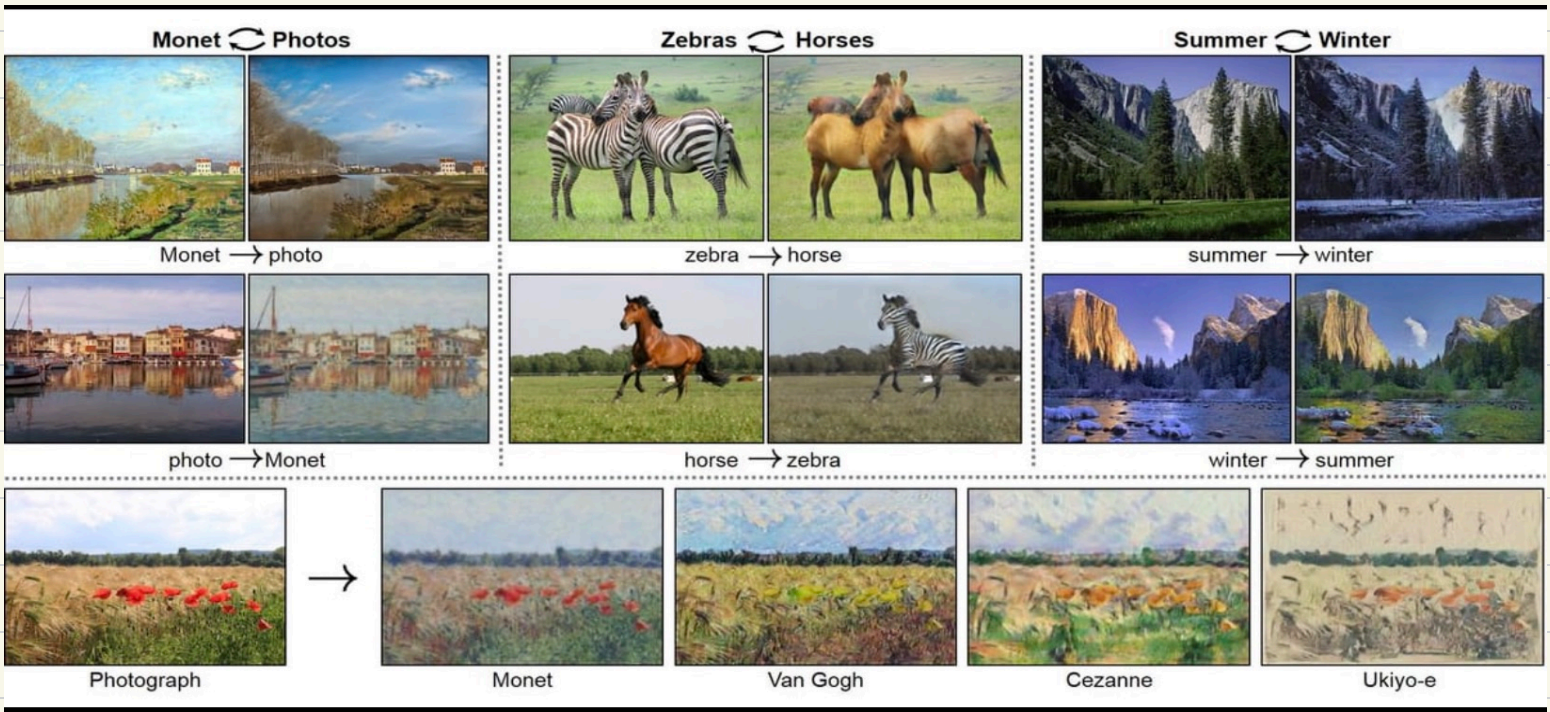
```code
Generate data that looks real
```

Step 7 — Repeat Training

Steps are repeated many times until:

```code
Generated data ≈ Real data
```

At this stage the discriminator **cannot easily distinguish fake data from real data**.



Latest Examples of GANs

1. AI Face Generation

GANs can generate **realistic human faces that do not exist in real life**.

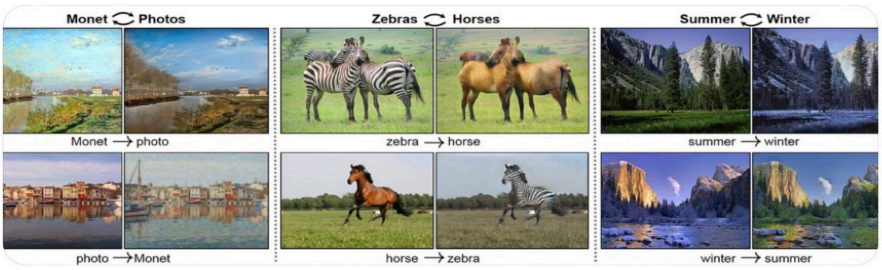
Example:

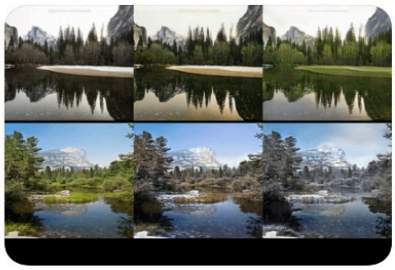
- The famous website **thispersondoesnotexist** generates a **new AI face every refresh**.
- These faces are created using **StyleGAN** developed by **NVIDIA**.

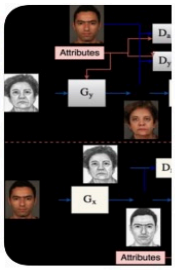
Use cases:

- virtual influencers
- gaming characters
- movie production

2. Image-to-Image Translation







GANs can convert **one type of image into another**.

Examples:

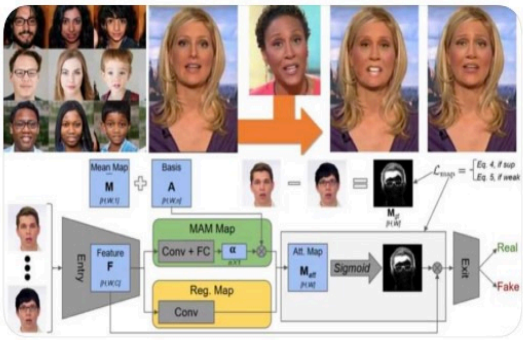
- horse → zebra
- sketch → realistic photo
- summer → winter scene

This is done using **CycleGAN**.

Use cases:

- film VFX
- game environment design
- photo editing

3. Deepfake Video Generation



GANs can generate **realistic face swaps in videos**.

Example uses:

- movies and VFX
- digital actors
- entertainment industry

But it also raises **ethical and security concerns**.

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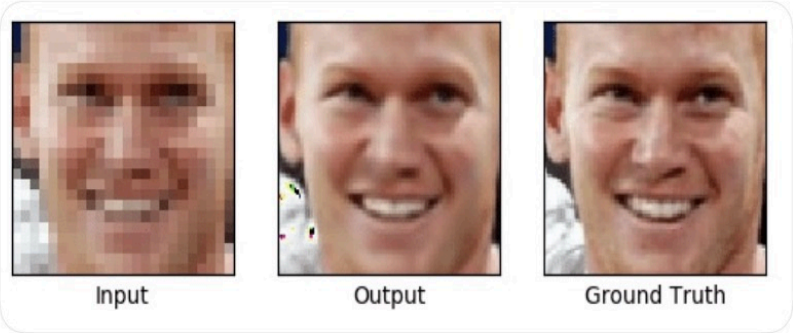
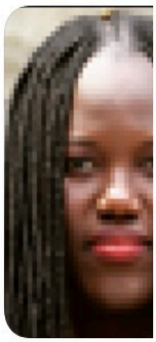
4. Super-Resolution (Image Enhancement)





GANs can convert **low-resolution images into high-resolution images**.

Example:

```text
Blurry Image → GAN → High-Resolution Image
```

Used in:

- satellite images
- security cameras